KIRAN ACADEMY

JAVA & PYTHON PROGRAMMING TASK

TASK 02: CAB BOOKING CHATBOT — MENU, FARE & OTP RULES

Submission Email	Subject	Task Date
9011@thekiranacademy.com	Task 02 – Cab Booking Chatbot – 16/10/2026	16 October 2026

1. Problem Statement

Create a console-based **Cab Booking Chatbot** in both Java and Python. Java students must implement it in Java; Python students must implement the same requirements in Python. The program must interact with the user, repeatedly show the main menu using **while (true)** in Java or **while True** in Python, calculate a fare using predefined routes and cab rates, and enforce the OTP cancellation rule.

2. Required Program Flow

Step	Menu / Action	What it must do
1	Main menu	Display **1. Book Ride** and **2. Exit**. Keep showing this menu until the user selects Exit. Handle invalid choices without crashing.
2	Choose cab type	For Book Ride, display **1. Auto**, **2. Sedan**, and **3. SUV**. Read the user's choice and use the corresponding per-kilometre rate.
3	Choose route	Display the three predefined routes in the table below. The user selects a route number. Use its fixed source, destination, and distance; do not ask the user to type arbitrary locations or kilometres.
4	Calculate fare	Fare = selected cab's rate per km × selected route distance. Display cab type, source, destination, distance, rate, and total fare.
5	Cancellation before OTP	Before displaying/sharing the OTP, ask the user to confirm the booking: **1. Confirm & Share OTP** or **2. Cancel Ride**. If cancelled, show a cancellation message and return to the main menu. Do not share an OTP for a cancelled ride.
6	OTP and ride lock	If confirmed, generate or assign a 4-digit OTP and display it as shared with the user. Once the OTP has been shared, the ride **cannot be cancelled**. Display this rule clearly. Show a booking-confirmed message and return to the main menu.
7	Exit	When the user selects 2 on the main menu, display a goodbye message and stop the loop/program. Any invalid menu choice should display an error message and show the menu again.

3. Predefined Routes and Cab Rates

Route No.	Source	Destination	Fixed Distance
1	Pune Station	Shivajinagar	5 km
2	Swargate	Kothrud	8 km
3	Hinjawadi	Wakad	10 km

Cab Type	Rate per km	Example for 8 km
Auto	■12/km	8 × 12 = ■96
Sedan	■18/km	8 × 18 = ■144
SUV	■25/km	8 × 25 = ■200

4. Sample Test Scenario

Input choices: 1 (Book Ride) → 2 (Sedan) → 3 (Hinjawadi to Wakad) → 1 (Confirm & Share OTP).

Expected calculation: Distance = 10 km; Sedan rate = ■18/km; Fare = 10 × 18 = ■180. The program shares a 4-digit OTP only after confirmation. After sharing the OTP, cancellation is not allowed.

5. Expected Output Behaviour

Your exact wording may differ, but the program should show the selected route, cab type, kilometres, per-kilometre rate, and fare; offer cancellation before OTP sharing; share an OTP only after confirmation; explain that cancellation is locked after OTP sharing; and keep returning to the main menu until Exit is selected.

6. Mandatory Rules

• Use user input: **Scanner** in Java and **input()** in Python.
• Use an infinite loop: **while (true)** in Java or **while True** in Python; exit using **break** or an equivalent clean loop condition.
• Use conditional statements for menu and cab choices. Store the route distances and cab rates as predefined values.
• Do not allow cancellation after OTP has been shared. Cancellation must be possible only before OTP sharing.
• Validate menu choices and avoid an endless loop for invalid input. Keep the program beginner-friendly; no database, API, GUI, or external library is required.

7. Submission Instructions — Mandatory

• **Student name:** Include your full name in the email body.
• **GitHub:** Upload your source code to your GitHub repository. Java students submit the Java program; Python students submit the Python program. You may include both if you are practising both languages.
• **Send to:** 9011@thekiranacademy.com
• **Subject:** Task 02 – Cab Booking Chatbot – 16/10/2026
• Do not attach source files to the email; submit the GitHub repository URL only.
• **Task date:** 16 October 2026.

Email body example: Hello, I am [Your Full Name]. I have completed Task 02 – Cab Booking Chatbot in [Java/Python]. GitHub Repository: [Paste repository URL]